AHP–TOPSIS Multi-Criteria Decision Visibility Engine

Algorithmic Specification, Mathematical Formulation, and Implementation in Agrivive

Project: Agrivive Marketplace Capstone | **Repository:** backend/src/utils/ahp-topsis | **Date:** October 2026
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Abstract & Context: The Weighted Surplus Visibility Algorithm in Agrivive is a multi-criteria decision-making method that determines the visibility level and display priority of active surplus vegetable listings. It uses the Analytic Hierarchy Process (AHP) to determine the importance of each criterion and the Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) to calculate scores and rank listings. AHP provides a structured way to assign criterion weights and check the consistency of comparisons, while TOPSIS evaluates listings based on their distance from the ideal best and worst results [1], [2]. These methods have also been applied in agricultural management, land suitability assessment, and precision farming [3]–[6].

The algorithm evaluates each eligible listing using four criteria: freshness based on posting time, stock availability, seller reliability, and historical surplus recurrence. First, AHP establishes the weights for these criteria through pairwise comparisons and checks their consistency. Next, TOPSIS normalizes the listing data, applies the weights, identifies the ideal best and worst results, and calculates a final score for each listing. Based on this score, Agrivive assigns one of three visibility levels: **Priority Boost** for scores of 0.60 and above, **Standard Exposure** for scores from 0.40 to below 0.60, and **Basic/Hold** for scores below 0.40.

This algorithm helps Agrivive prioritize listings using a consistent and explainable process rather than relying on a single factor or arbitrary ranking. By considering multiple criteria, it prevents one listing characteristic alone from determining the final priority and helps identify listings that may benefit from greater marketplace exposure [2]. Listings under Priority Boost may receive better feed placement, a visibility badge, and eligibility for promotional activities through n8n, while other listings remain accessible through regular marketplace discovery. This supports sellers in promoting their remaining vegetables and helps buyers discover available surplus products. However, the visibility score measures promotional priority only and does not determine the actual freshness, safety, or marketability of the vegetables.

1. System Architecture & Dual-Policy Implementation

Agrivive implements the ranking engine using a dual-policy architecture controlled by the server environment variable VISIBILITY_POLICY. This decoupling allows live A/B benchmarking and safe zero-downtime rollback between the baseline model and the new pilot decision engine. Under the baseline mode (visibility-v4), the system executes a linear Weighted Surplus Visibility (WVS) scoring formula: **V = 0.40P + 0.40Q + 0.20R**, where P represents elapsed posting age, Q is remaining inventory quantity, and R is previous listing cycle count. While functional, the baseline assigns fixed heuristic weights and inadvertently rewards aging produce. Under the pilot policy (visibility-ahptopsis-v1), the system activates the full AHP–TOPSIS multi-criteria pipeline implemented in backend/src/utils/ahp-topsis/index.ts, incorporating four rigorously normalized benefit criteria and matrix consistency verification.

2. Detailed Discussion of the Four Evaluated Criteria

Each active candidate listing is evaluated against four quantitative criteria. To establish geometric comparability across disparate physical units (hours, kilogram ratios, review stars, and cycle counters), every criterion is structured as a **benefit criterion** (where higher normalized values signify higher marketplace promotion priority) and clamped strictly within the closed interval [0, 1].

2.1 Criterion 1: Freshness (C■)

Operational Definition & Rationale: Freshness addresses the rapid post-harvest degradation inherent to tropical horticultural crops. The criterion measures the elapsed chronological time between listing publication and evaluation. Because quality deterioration in leafy and fruit vegetables begins immediately post-harvest, Agrivive establishes a 12-hour evaluation horizon. If t_eval is the query timestamp and t_pub,i is the listing publication time, the elapsed age in hours is calculated as age_i = (t_eval - t_pub,i) / 3600. The normalized freshness metric x_i1 is:

$$x_{i1} = \text{clamp}(1 - age_i / 12, 0, 1) = \max(0, \min(1, 1 - age_i / 12))$$

Equation 1: Freshness Benefit Metric (12-Hour Linear Decay Horizon)

Platform Usage: This metric assigns a perfect score of 1.0 to freshly published listings, linearly decaying to 0.0 once elapsed age reaches or exceeds 12 hours. This operational priority ensures newly harvested surplus produce receives immediate customer visibility and accelerated turnover before physiological decay occurs.

2.2 Criterion 2: Stock Availability (C■)

Operational Definition & Rationale: Stock Availability measures the proportion of unreserved produce volume remaining in the active sales batch relative to the initial quantity declared when the listing cycle was created. Letting q_current,i represent unreserved inventory and q_original,i represent starting batch inventory, x_i2 is computed as:

$$x_{i2} = \text{clamp}(q_{\text{current, i}} / q_{\text{original, i}}, 0, 1) = \max(0, \min(1, q_{\text{current, i}} / q_{\text{original, i}}))$$

Equation 2: Remaining Surplus Stock Availability Ratio

Platform Usage: This criterion assists sellers in clearing large remaining surpluses by prioritizing batches with substantial unreserved inventory. Conversely, batches with only residual quantities near zero score lower, preventing nearly exhausted listings from dominating marketplace discovery.

2.3 Criterion 3: Seller Reliability (C■)

Operational Definition & Rationale: Seller Reliability incorporates vendor trust and verified fulfillment into the multi-criteria model. Letting R_i in [1, 5] represent the seller's cumulative mean star rating derived from verified buyer post-pickup evaluations, the metric x_i3 is normalized against the 5-star ceiling:

$$x_{i3} = \text{clamp}(R_i / 5, 0, 1), \quad \text{with default fallback } x_{i3} = 0.50 \text{ when no review history exists}$$

Equation 3: Normalized Seller Reliability Rating with Cold-Start Baseline

Platform Usage: To address the cold-start barrier and prevent structural discrimination against newly registered farmers and vendors, the engine applies an unpenalized neutral baseline of 0.50 (equivalent to 2.5 stars) whenever review records are absent. This safeguards consumer confidence while preserving open access for new market participants.

2.4 Criterion 4: Surplus Recurrence (C■)

Operational Definition & Rationale: Recurrence quantifies long-term supply consistency and vendor engagement for a specific vegetable commodity. Linked through Agrivive's standardized botanical catalog identifier and selling unit (e.g., Sitaw sold per kilogram versus Sitaw per bundle), the engine tallies the seller's prior completed zero-stock replenishment cycles (prior cycles_i):

$$x_{i4} = \text{clamp}(\text{prior cycles}_i / 3, 0, 1) = \max(0, \min(1, \text{prior cycles}_i / 3))$$

Equation 4: Standardized Catalog Recurrence Factor (3-Cycle Saturation)

Platform Usage: This metric rewards vendors who demonstrate sustained agricultural supply reliability, reaching maximum saturation (1.0) after three completed stock turnover cycles. It differentiates active, dependable marketplace producers from isolated or irregular sellers.

Criterion	Symbol	Physical Source	Normalized Domain	AHP Derived Weight
Freshness	C■	Elapsed hours since publication	[0, 1] (12-hr decay)	**0.400 (40%)**
Stock Availability	C■	Current vs. original batch qty	[0, 1] (proportion)	**0.300 (30%)**
Seller Reliability	C■	Verified order review rating	[0, 1] (0.50 default)	**0.200 (20%)**
Surplus Recurrence	C■	Prior completed catalog cycles	[0, 1] (3-cycle cap)	**0.100 (10%)**

3. Step-by-Step Mathematical Workflow & Formulas

The algorithmic engine executes in two distinct phases: Phase 1 derives criterion weights using the Analytic Hierarchy Process (AHP) [1] and verifies matrix consistency; Phase 2 executes the Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) [2] across all active listings to produce a normalized closeness index.

3.1 Phase 1: Analytic Hierarchy Process (AHP) Weight Derivation

Step 1 — Pairwise Comparison Matrix Construction: Domain agricultural criteria priorities are formalized into an m x m reciprocal comparison matrix A = [a_jk] (where m = 4) using Saaty's 1–9 fundamental scale [1]. The elements satisfy reciprocal condition a_kj = 1 / a_jk and diagonal identity a_jj = 1.0:

Criteria Comparison	C■ (Fresh)	C■ (Stock)	C■ (Reliab)	C■ (Recurr)
C■: Freshness	1.000	1.333	2.000	4.000
C■: Stock Availability	0.750	1.000	1.500	3.000
C■: Seller Reliability	0.500	0.667	1.000	2.000
C■: Recurrence	0.250	0.333	0.500	1.000

Step 2 & 3 — Geometric Mean and Weight Vector Computation: To avoid rank reversal and maintain reciprocal symmetry, the engine computes the unnormalized geometric mean g_j for each row j, followed by normalization across total sum:

$$g_j = \left(\prod_{k=1}^{m} a_{jk} \right)^{1/m} \quad \Rightarrow \quad w_j = g_j / \sum_{r=1}^{m} g_r$$

Equation 5: Geometric Mean (Logarithmic Least Squares) and Normalized Weight Vector

Evaluating Equation 5 across Matrix A yields unnormalized geometric means g_1 = 1.807, g_2 = 1.355, g_3 = 0.904, and g_4 = 0.452 (summing to 4.518). Normalization produces the definitive weight vector: **w = [0.400, 0.300, 0.200, 0.100]■**,

satisfying the unity axiom ($\sum w_j = 1.000$).

Step 4 — Mathematical Consistency Verification: To verify that subjective pairwise judgments do not contain intransitive contradictions (e.g., A > B and B > C but C > A), eigenvalue analysis computes the weighted sum vector $s_j = \sum a_{jk} w_k$ and individual eigenvalue ratios $\lambda_j = s_j / w_j$. The maximum eigenvalue λ_{max}, Consistency Index (CI), and Consistency Ratio (CR) are evaluated against Saaty's empirical Random Index (RI_m = 0.90 for m = 4) [1]:

$$\lambda_{max} = (1/m) \sum_{j=1}^{m} (s_j / w_j), \quad CI = (\lambda_{max} - m) / (m - 1), \quad CR = CI / RI_m$$

Equation 6: Maximum Eigenvalue, Consistency Index (CI), and Consistency Ratio (CR)

For the Agrivive comparison matrix, λ_{max} evaluates to exactly 4.000, yielding CI = 0.000 and CR ≈ 0.000. Because CR is substantially below the established threshold of 0.10, the matrix achieves complete mathematical transitivity.

3.2 Phase 2: TOPSIS Ranking and Closeness Computation

Step 5 & 6 — Decision Matrix Formation and Vector Normalization: For n active eligible listings, the decision matrix $X = [x_{ik}]_{(n \times 4)}$ aggregates the four normalized benefit criteria. To eliminate dimensional distortion and map all criteria onto a shared geometric scale, Euclidean vector normalization transforms each element into r_{ik}:

$$r_{ik} = x_{ik} / \sqrt{\left(\sum_{i=1}^{n} x_{ik}^2 \right)}$$

Equation 7: TOPSIS Vector Normalization

Step 7 — Weighted Normalized Decision Matrix Construction: Each normalized column is multiplied by its AHP weight w_k:

$$v_{ik} = w_k \times r_{ik}$$

Equation 8: Weighted Normalized Attribute Formulation

Step 8 — Identification of Positive and Negative Ideal Solutions: Because all four criteria are formulated as benefit metrics, the Positive Ideal Solution (A■) represents the maximum column elements, while the Negative Ideal Solution (A■) represents the minimum column bounds:

$$A^+ = (\max_i v_{i1}, \max_i v_{i2}, \max_i v_{i3}, \max_i v_{i4}) = (v_1^+, v_2^+, v_3^+, v_4^+)$$
$$A^- = (\min_i v_{i1}, \min_i v_{i2}, \min_i v_{i3}, \min_i v_{i4}) = (v_1^-, v_2^-, v_3^-, v_4^-)$$

Equation 9: Positive-Ideal (A■) and Negative-Ideal (A■) Benchmark Coordinates

Step 9 & 10 — Euclidean Distance and Relative Closeness Coefficient Computation: The Euclidean separation of each listing from the ideal-best (D_i■) and ideal-worst (D_i■) is measured. The final closeness coefficient C_i^* determines overall promotional priority [2]:

$$D_i^+ = \sqrt{\left(\sum_{k=1}^{4} (v_{ik} - v_k^+)^2 \right)}, \quad D_i^- = \sqrt{\left(\sum_{k=1}^{4} (v_{ik} - v_k^-)^2 \right)}$$

$$C_i^* = D_i^- / (D_i^+ + D_i^-), \quad \text{where } 0 \le C_i^* \le 1$$

Equation 10: Multidimensional Euclidean Distances and Relative Closeness Coefficient

Step 11 — Ordering, Deterministic Tie-Breaking, and Tier Assignment: Candidate listings are sorted in descending order of C_i^* ($C_{(1)}^* \ge C_{(2)}^* \ge \dots \ge C_{(n)}^*$). In the event of identical numerical closeness coefficients, pagination stability is maintained through deterministic tie-breaking: listings are sorted by publication timestamp descending (publishedAt), followed by lexicographical sorting of unique listing identifiers (id). Finally, continuous scores are partitioned into discrete

marketplace tiers:

$$\text{VisibilityTier}(C_i^*) = \text{Priority Boost } (C_i^* \geq 0.60) \mid \text{Standard Exposure } (0.40 \leq C_i^* < 0.60) \mid \text{Basic / Hold } (C_i^* < 0.40)$$

Equation 11: Marketplace Exposure Tier Classification Boundaries

Visibility Tier	Score Domain	Marketplace Treatment	Automated Promotion Impact
Priority Boost	$C_i^* \geq 0.60$	Top search feed placement & distinctive badge	Eligible for automated n8n official social dispatches
Standard Exposure	$0.40 \leq C_i^* < 0.60$	Standard catalog discovery feed placement	Available through organic search; no external boost
Basic / Hold	$C_i^* < 0.40$	Direct category search and store view only	Retained in catalog; not highlighted in public feed

4. Operational Value to Agrivive and Stakeholders

The AHP–TOPSIS multi-criteria engine provides four foundational advantages to the platform:

1. Cold-Start Viability: Unlike machine-learning recommenders that depend on extensive historical purchase records, AHP–TOPSIS functions deterministically using day-one operational data, resolving cold-start data sparsity.

2. Explainable Agricultural Alignment: Pairing 40% harvest freshness with 30% surplus stock ratio directly accelerates the turnover of high-volume perishable produce before quality degradation occurs, directly addressing local food loss.

3. Objective Weight Justification: Moving away from guessed heuristic percentages to an AHP comparison matrix provides empirical consistency proof (CR ≈ 0.000 < 0.10), giving the academic capstone rigorous scientific defensibility [1], [3].

4. Safe Deployment & Reproducibility: Controlled behind the VISIBILITY_POLICY environment flag and verified through the companion simulation workbook (outputs/agrivive-ahp-topsis-simulation.xlsx), the engine allows empirical benchmark evaluation against legacy policies without threatening live marketplace operations.

5. References and Academic Literature

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